Self-Balancing Dictionary Learning for Relaxed Collaborative Representation of Hyperspectral Image Classification

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***Abstract*— Supervised dictionary learning and representation (S-DLR) learning framework has demonstrated its superiority for hyperspectral image classification. Relaxed collaborative representation (RCR) has also been acknowledged as an effective method in balancing the similarity and difference between features. In this article, a new dictionary learning method is introduced to balance the discrimination and reconstruction of training samples. In the dictionary learning stage, two new indicators are designed to measure the discriminability of items and can be improved by optimizing coding coefficients. The imposed l_2-norm between the independent item and the mean of class-specific samples constrains the similarity, and the calculated weights measure the difference. In the label determination stage, considering that the residuals of RCR are adversely affected by the significant difference of features, a new classification approach is introduced. Class labels are assigned without calculating the reconstruction errors but calculating the levels of comprehensive contribution from all training samples instead. Since the effectiveness will be degraded when dealing with more complex circumstances, a region-based version is further introduced. It can further improve the discrimination of dictionary items due to the reduced categories in each subimage and reduce the computation cost. The experimental results on several hyperspectral datasets demonstrate that our methods can effectively improve the classification performance.**

***Index Terms*— Dictionary learning, hyperspectral classification, relaxed collaborative representation (RCR).**

I. Introduction

HYPERSPECTRAL imaging technique is of great interest in real applications [1], [2], [3]. Its high spectral resolution can help differentiate ground objects, which brings more possibilities for a broad of applications such as target detection [4], classification [5], and environment monitoring [6]. Nevertheless, the existence of Hughes phenomenon [7] limits the exertion of these advantages in hyperspectral image classification [8], [9], [10]. It is necessary to design effective strategies to address the challenges [11], [12], [13], [14], [15], [16], [17].

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A. Representation Learning

Representation learning proposed in [18] is an advantageous model, which has contributed to achieve considerable performance in various fields such as [19], [20], and [21]. Willett et al. [19] employed the sparse representation classification (SRC) model for hyperspectral image classification since it is clearly interpretable and without complex procedures for parameter optimization. Zhang et al. [22] changed the l_1-norm into l_2-norm and proved that the superiority of the representation algorithm lies in collaboration rather than sparsity. Then, they proposed collaborative representation (CR). As for these models, the training samples are used directly to build a dictionary, but it may be redundant [19]. It is necessary to learn a more suitable dictionary from selected training samples. Jointing supervised dictionary learning and representation (S-DLR) plays an important role in this regard according to [23].

B. Supervised Dictionary Learning (S-DL)

As for dictionary construction, the recent strategies can be roughly categorized into three groups: sharedness-based, class-based, and the combination of both learning methods.

1) Shared Dictionary Learning Method: It learns a fixed-size dictionary by capturing training samples from all classes. Elad and Aharon [24] took the similarity between clustering members into account and proposed K-means-based singular value decomposition (K-SVD), which can be regarded as a basic benchmark algorithm [25], [26], [27], [28], [29]. In order to enhance the discrimination of the dictionary in K-SVD, Zhang and Li [25] constructed a new objective function by jointly refining the dictionary and classifier parameters. The discriminative K-SVD (D-KSVD) method is built on the comprehensive consideration. Besides, considering the autocorrelation of training samples, Jiang et al. [26] incorporated class labels into the construction of a discriminative coding error term and proposed label-consistent K-SVD (LC-KSVD). Tang et al. [30] showed that it is also effective in hyperspectral image classification. Considering the insufficient training samples, Xie et al. [32] divided a dictionary into positive and negative parts by using a mutual exclusion term. The mutually exclusive KSVD (ME-KSVD) can take the interclass difference and intraclass consistency into account. Apart from these methods which mainly focus on spectra, spatial information is also noted in the hyperspectral image

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classification. Soltani-Farani et al. [31] segmented a hyperspectral image into several spatially adjacent contextual groups and proposed spatial-aware dictionary learning (SADL). In the first case, the shared dictionary learning captures common features (i.e., spectral and spatial features) well but loses the unique characteristics to some extent.

2) Class-Based Dictionary Learning Method: This type of method designs a class-specific strategy to capture the common characteristics of training samples. It learns a subdictionary per class and then composes the partial into a final one. In view of the characteristics of hyperspectral data in which the samples come from the same class but present different spectra, it is considered to be a proper and effective method. Yang et al. [33] not only focused on the independent dictionary items constraint but also considered the coding coefficients constraint, and the double constraint method is a Fisher discrimination dictionary learning (FDDL) model. Inspired by [33], Zongze et al. [34] joined the FDDL and majority voting scheme to improve the discrimination of samples in each class. Based on the advantage of K-SVD in capturing the shared information, Tu et al. [35] designed a global discriminative constraint and proposed discriminative subdictionary learning (DSDL). It minimized the influence of other classes as much as possible. Akhtar and Mian [36] learned subdictionaries for each class and used Bernoulli distributions [37] to measure the frequency of their contribution in representation. The distributions represent the energy of discrimination of training samples from classes. In the second case, the class-specific method effectively extracts discriminative features by imposing constraints on specific factors. However, the common characteristics may also be ignored.

3) Commonality and Particularity Dictionary Learning Method: In this circumstance, the dictionary is constructed by two contributors: commonality and particularity. The former contributes to coefficients representation and the latter extracts discriminative features. Li et al. [38] proposed a framework for spectral feature extraction. The devised shared constraints are imposed on subdictionaries to consider the commonality. Meanwhile, the results of spectral convolution are used to extract the discriminative information. Wei et al. [39] added up weighted sparsity and triplet relation preserving terms into an objective function to constrain the similarity among pixels. By spectral segmentation, the discriminative between-class features can be captured. Furthermore, the block-based spectral information motivates to generate discriminative sparse coding coefficients with the imposed discriminability promoting term. In the third case, the commonality and particularity dictionary learning method comprehensively considered the reconstruction and discrimination of interclass and intraclass features. However, the construction of a universal and effective dictionary is in great uncertainty when dealing with different hyperspectral images.

C. Classification Approach

In terms of the recoding after dictionary learning, there are mainly two forms. The first method is regarding the learned dictionary as a new input of a nonlinear classifier such as support vector machine (SVM) [31], [40] and extreme learning machine (ELM) [38], [41]. In this circumstance, the principle for determining class labels depends on the specific classifier. The second method concentrates on the minimization of reconstruction errors by constructing specific constraints. Generally, the imposed constraints are with respect to the process of dictionary learning. Su et al. [42] extracted multifeatures to construct diverse dictionaries and generated the coding coefficients by the CR model. Tang et al. [30] utilized a sparse representation model to estimate a representation coefficient matrix after obtaining a learned dictionary from LC-KSVD. Besides, Fang et al. [43] incorporated the spatial information into representation and established multiscale adaptive sparse representation (MASR). To consider spatial information simultaneously, Xie et al. [32] input the learned dictionary containing spectral features into MASR. Recently, relaxed collaborative representation (RCR) [44] considered similarity and difference between features as effective characteristics when constructing coding coefficients matrices. In this model, the reconstruction errors include residuals from different features. It has demonstrated great superiority in balancing similarity and difference between bands [45]. From a functional point of view, it can be considered as an excellent classifier in balancing similarity and difference. This advantage exactly coincides with the requirement of distinguishing homogeneous objects in hyperspectral data.

In summary, problems in S-DLR can be in two categories. For dictionary learning, the first to be solved is balancing the discriminative and shared energy. The second one is keeping its flexibility as well as adaptability to diverse samples when dealing with various hyperspectral images. For coding coefficients, the crucial problem that needs to be solved is designing an effective framework of representation and determination of class labels. It needs to vigorously capture the potential discriminative information contained in a dictionary.

D. Proposed Contribution

In this article, to overcome the aforementioned drawbacks, a self-balancing dictionary learning method based on RCR (SBDL_RCR) is introduced. In the dictionary learning stage, it is required to learn a subdictionary for each class. The original dictionary will be represented by itself first. Two indicators are designed to measure the discrimination. Then, the above self-representation coefficients are optimized to help to improve the discrimination of the items. Besides, a balance program is imposed to maintain the reconstructive power. Thus, a comprehensive dictionary, which considers both discrimination and reconstruction, is formed. In the label determination stage, a new framework of contribution indicator is proposed for the first time. All samples will be employed to participate in classifying. Besides, a region-based version (RSBDL_RCR) is proposed to improve the classification performance and reduce the computation cost. The specific contributions are given as follows.

1) To evaluate the discrimination of each item, two new evaluation indicators, self-contribution (SC) level and

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average-contribution (AC) level, are designed and can be improved by optimizing the self-representation coefficients of the initial dictionary.
2) A novel balance program is imposed to consider the discriminative and reconstructive power simultaneously. The weights and l_2 norm terms are employed to play a constraint role, while the balancing parameter can control the similarity of samples.
3) Considering the disturbing impact of residuals from RCR, a new classification approach is introduced. It consists of a pair of education indicators, SC, and AC levels. Due to its characteristic of recruiting all samples, the flexibility of the learned dictionary is also improved.
4) To achieve better classification performance and reduce the computational cost, a region-based version is further proposed. The categories can be reduced in each subimage and the discrimination of items can be further improved.

The remainder of this article is organized as follows. The related work, including RCR, dictionary learning, and image segmentation, are briefly introduced in Section II. Section III describes the proposed methods in detail, including an original and a region-based version. The experimental results are presented and discussed in Section IV. Section V summarizes our work and presents the potential for further work.

II. Related Work

A. Relaxed CR

The RCR model [44] mainly delves into the problem of various intensities between diverse features that participate in describing the samples. As for hyperspectral image, its hundreds of bands bring considerable information indeed but are also in low feature diversity. Multiple features [46], [47], [48] are therefore introduced and RCR is proposed to weight the increased diversity according to the mutual distance.

In the coding stage, a regularization constraint between coefficients obtained from each feature and the mean of the coefficients ensures the small variance. The weights are optimized according to the coding coefficients to measure the difference between features. The objective function of RCR is

$$\arg\min_{\alpha_k} \sum_{k=1}^{K} ||y_k - D_k\alpha_k||_2^2 + \lambda||\alpha_k||_2^2 + \tau\omega_k||\alpha_k - \bar{\alpha}||_2^2 \quad (1)$$

where K is the number of features. λ and τ are scale parameters that balance the reconstruction error, the sparsity of coefficients, and the regularized power. The mean of coefficients can be calculated as

$$\bar{\alpha} = \omega_k\alpha_k \Big/ \sum_{k=1}^{K} \omega_k. \quad (2)$$

The objective function (1) can have a closed form if ω is known. Nevertheless, there is weak prior information between involved multiple features since their unknown relationships. Thus, the regularization of entropy is imposed on the third additional coefficient constraint term in (1), and the specific constraint function can be

$$\min_{\omega_k} \sum_{k=1}^{K} \tau\omega_k||\alpha_k - \bar{\alpha}||_2^2 + \gamma\,\omega_k \ln\omega_k \quad (3)$$

where γ is a Lagrange multiplier. Then, weights can be updated by

$$\omega_k = \exp\left(-1 - \tau||\alpha_k - \bar{\alpha}||_2^2/\gamma\right). \quad (4)$$

The coding coefficients can be obtained by

$$\alpha_k = \alpha_{0,k} + \tau\frac{\omega_k}{\sum_{\eta=1}^{K}\omega_\eta} U_k V \sum_{\eta=1}^{K} \omega_\eta\alpha_{0,\eta} \quad (5)$$

$$U_k = \left(D_k^T D_k + I(\lambda + \tau\omega_k)\right)^{-1} \quad (6)$$

$$\alpha_{0,k} = U_k D_k^T y_k \quad (7)$$

$$V = \left(I - \sum_{\eta=1}^{K} \tilde{\omega}_\eta U_\eta\right)^{-1} \quad (8)$$

$$\tilde{\omega}_\eta = \frac{\tau\omega_\eta^2}{\sum_{k=1}^{K}\omega_k}. \quad (9)$$

In the classification stage, RCR combines reconstruction errors from all features according to their weights updated during the process of coding. The final classification can be achieved by

$$r_i = \sum_{k=1}^{K} \omega_k\left|\left|y_k - D_k^i\alpha_k^i\right|\right|_2^2. \quad (10)$$

B. Dictionary Learning

The model for joint dictionary and representation learning has been applied in many fields, such as denoising [49], image compression [50], image super-resolution [51], and classification [36]. According to S-DL, the utilization of label information can motivate the improvement of accuracy and has led to great results in a variety of classification tasks [52], [53]. In general, a testing sample can be represented linearly over a dictionary, which consists of training samples without actual training. It may result in an inefficient dictionary of the influence of noise. To obtain a dictionary that is as effective as possible and less affected by noise, the mathematical expression of dictionary learning is given as follows [54]:

$$\arg\min_{(D,A)} |\left(||Y - DA||_F^2 + \lambda_1||A||_{l_p}\right) + \lambda_2 f_A(A) + \lambda_3 f_D(D) \quad (11)$$

where λ_1, λ_2, and λ_3 are regularization parameters, f_A and f_D are task-driven constraints, and l_p represents the norm constraint.

According to the way of working out coding coefficients, there are two circumstances. In the first case, the dictionary and coding matrix can be solved iteratively and alternately since they are both unknown. One should be fixed while updating the other in each iteration. The other is

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Algorithm 1 SBDL_RCR

Input: Training samples $D^k(k = 1, 2, \ldots, K)$, testing samples $Y^k(k = 1, 2, \ldots, K)$, regularization parameter λ, balance parameter β, constraint parameter $\tau and \gamma$, maximum number of iterations $iters$

1. Dictionary Self-representation Stage:
For k=1: K
  initial dictionary: $D^{(0.k)} = D^k$
  self-representation coefficients by (1): $A^{(0.k)} = A^k$
  $D^{(*.k)} = D^{(0.k)}$
  $A^{(*.k)} = A^{(0.k)}$
end For
2. Self-representation Coefficients Optimization Stage:
For k=1: K
  for i=1: N
    for j=1: N
      computing $\frac{\partial F}{\partial A_{ij}}\left(D^{(*,k)}, A^{(*,k)}\right)$ by (16)
      while convergence **do**
        updating A_{ij}^* by iteratively moving in descent direction $-\frac{\partial F}{\partial A_{ij}}$ using (17) - (19)
      end while
    end for
  end for
end For
3. Dictionary Self-balancing Stage:
For k=1: K
  for c=1: L
    while convergence **do**
      computing ω by (23)
      updating D_c by (21)
    end while
  end for
end For
4. Classification Stage:
Determining the label by (25) - (27)
Output: $class(Y)$

Algorithm 2 RSBDL_RCR

Input: Training samples $D^k(k = 1, 2, \ldots, K)$, testing samples $Y^k(k = 1, 2, \ldots, K)$, regularization parameter λ, balancing parameter β, constraint parameter τ *and* γ, maximum number of iterations $iters$, the number of super-pixels nBs
For ns=1: nBs
  1. Dictionary Self-representation Stage
  2. Self-representation Coefficients Optimization Stage
  3. Dictionary Self-balancing Stage
  4. Classification Stage
  5. Determining the label by (25) - (27)
end For
Output: $class(Y)$

the post-computation for calculating the coding coefficients in which circumstance the dictionary can be obtained first. Due to the high complexity when calculating the coefficients for RCR [45] in hyperspectral classification, the alternative updating is not favorable in this circumstance.

C. Image Segmentation

As a significant preprocessing procedure in the field of computer vision, image segmentation shows its advantages in reducing the complexity [55] and plays a central role in a broad range of fields [55], [56], [57]. It involves partitioning the input image into several subimages according to some common characteristics [58]. In particular, hyperspectral images are characterized by the rich information of underlying materials, which can help to obtain accurate segmentation. The segmented hyperspectral images improve the classification efficiency to a large extent. Hence, in the classification of hyperspectral images with high complexity, it can be reduced by image segmentation first.

The traditional methods of segmentation include K-means clustering [59], conditional and Markov random fields [60], and so on. They have higher computational cost than superpixel segmentation [61], [62]. Superpixels [62] obtain concise subgraphs by aggregating some pixels with similar characteristics to form a homogeneous region. The subgraphs are believed to have low complexity and less data redundancy, which can reduce the cost of the subsequent tasks. Meanwhile, it creates favorable conditions to improve the performance of the algorithm.

According to [62], entropy rate superpixel segmentation (ERS) shows great performance in all the standard evaluation metrics and segments the image effectively. It is composed of two effective components. The entropy rate of a random walk on a graph encourages to form clusters with homogeneity and compactness, while the balancing term can obtain subgraphs with similar sizes. As for an input image, it can be regarded as a graph $G = (V, E)$, where V and E represent the collection of vertices and edges, respectively. Graph construction can be achieved by

$$\max_A = \mathcal{H}(A) + \lambda \mathcal{B}(A) \quad \text{s.t. } A \subseteq E. \tag{12}$$

Here, $\lambda \geq 0$ is a scale parameter for adjusting the balancing term according to specific tasks and A represents the collection of selected edges. Liu et al. [62] introduced more information about the principles and details, including the detailed definitions of (12).

In this model, the superpixels are generated hierarchically, as shown in Figs. 1 and 2. The superpixels in lower levels will be merged into the higher ones, which is adaptive to various images.

III. Proposed Algorithm

In this article, a new dictionary learning method is proposed to balance the difference and similarity between samples in each class. First, the input dictionary can be considered as a testing set and the item is treated separately. In this self-representation stage, the discrimination of each item will be measured first by the new proposed SC level and AC level. Then, a gradient descent method is used to optimize the self-representation coding coefficients to obtain the local optimum. Thus, the discrimination of each item can be improved by enhancing self-representation power and weakening other

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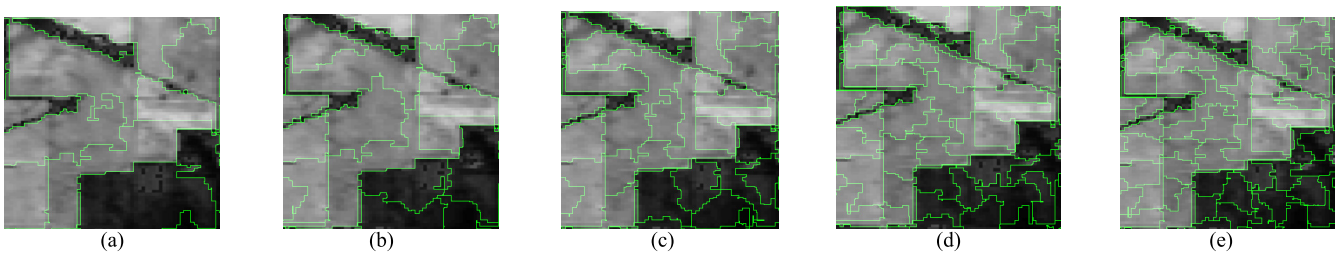


Fig. 1. Segmented results of AVIRIS data under different scales. (a) Number of superpixels = 50. (b) Number of superpixels = 100. (c) Number of superpixels = 200. (d) Number of superpixels = 300. (e) Number of superpixels = 400.

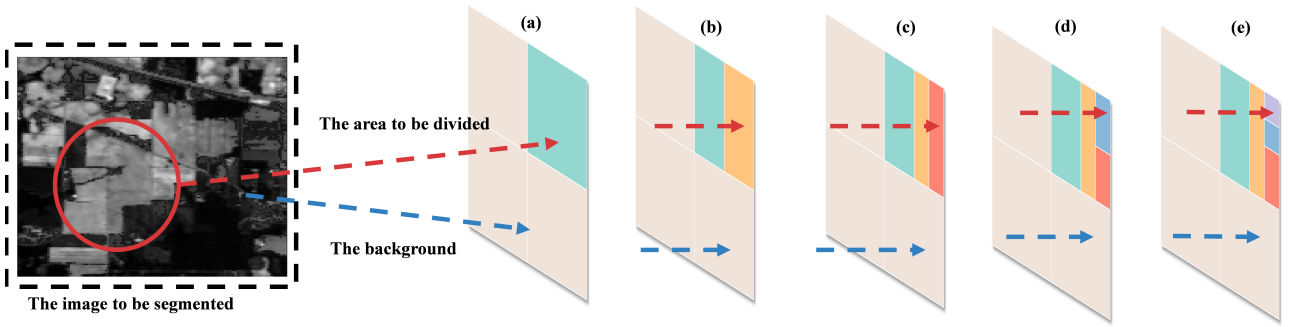


Fig. 2. Indication process of ERS. (Taking the process of an area to be segmented as an example. The area to be divided will obtain a hierarchical segmentation and (a)–(e) represent this process).

influential interference. Meanwhile, taking the reconstruction power into account, the balancing program is applied to items with improved discriminative power.

A. SBDL_RCR

As for supervised representation learning, the training samples of each class are characterized by high similarity but low identifiability. Nevertheless, a great dictionary should involve discriminative items to improve its efficiency. The reconstructive power cannot be ignored as well. Thus, a new dictionary learning method, which concentrates on the balance between discrimination and reconstruction, is proposed. Considering the high complexity of RCR, the coding coefficients are going to be optimized first and the dictionary is post-calculated.

To begin with, select equal samples from each class. In consideration of the employed multifeatures (i.e., extended morphological attribute profiles (EMAP), local binary pattern (LBP), and Gabor filter (Gabor) features), the number of final dictionaries will be in line with the multiple features.

In the dictionary self-representation stage, the objective function for each feature is expressed as

$$F1 = \arg\min_{(D,A)} ||D^k A^k - X^k||_F^2 + \lambda \sum_{i=1}^{N}\sum_{j=1}^{N} \left|\left|A_{ij}^k\right|\right|_2^2 + \tau \sum_{i=1}^{N}\sum_{j=1}^{N} \left|\left|A_{ij}^k - \tilde{A}_{ij}\right|\right|_2^2. \quad (13)$$

Let $D = [D^1, D^k, \ldots, D^K]$ and $D^k \in R^{dk\times N}(k = 1, 2, \ldots, K)$, where N is the number of training samples and dk is the dimension of the kth feature. The coding coefficient matrices in the self-representation stage are defined as $A = [A^1, A^k, \ldots, A^K](k = 1, 2, \ldots, K)$, $A^k \in R^{N\times N}$. It is noticed that $X = D$ in this circumstance. Generally, self-representation coefficients of the original dictionary reflect the ability to distinguish similar classes. The contribution of an item itself should be much larger than that of the others. $\tilde{A}$ is defined as the constraint matrix to represent the ideal situation in which $\tilde{A}_{ij} = 1(i = j)$ and $\tilde{A}_{ij} = 0(i \neq j)$. The coding coefficients are calculated by RCR.

The proposed SC level to measure the discrimination quantitatively can be defined as

$$P_n = ||A_{n,N_c}||_1 / ||A_{n,N}||_1, \quad (n = 1, 2, \ldots, N), \quad (c = 1, 2, \ldots, L) \quad (14)$$

where $P \in R^{N\times L}$, n represents the nth sample, and L is the number of classes. Figs. 3 and 4 show the effectiveness of optimization. As for samples within class, SC is larger naturally, but the gap between the item and other interclass samples is more pronounced. SC levels of the items can approximately reach 1 after optimization. To consider the collaborative contribution of different classes, the AC level, which measures the importance of each class, is given as

$$Q_c = \left|\left|P_{Nc}^c\right|\right|_1 / \left|\left|P_N^c\right|\right|_1 \quad (15)$$

where $Q \in R^{1\times L}$. The initial SC and AC levels can be calculated first before the next optimization step.

In the optimization stage, the dictionary self-representation coefficients are updated by the gradient descent method to get the local optimum. According to (13), the gradient can be calculated by

$$\frac{\partial F1}{\partial A_{ij}} = 2d_{.j}^T D A_{.j} - 2d_{.j}^T X_{.j} + 2\lambda \frac{A_{ij}}{|A_{ij}|} + 2\tau \frac{A_{ij} - \tilde{A}_{ij}}{|A_{ij}|}. \quad (16)$$

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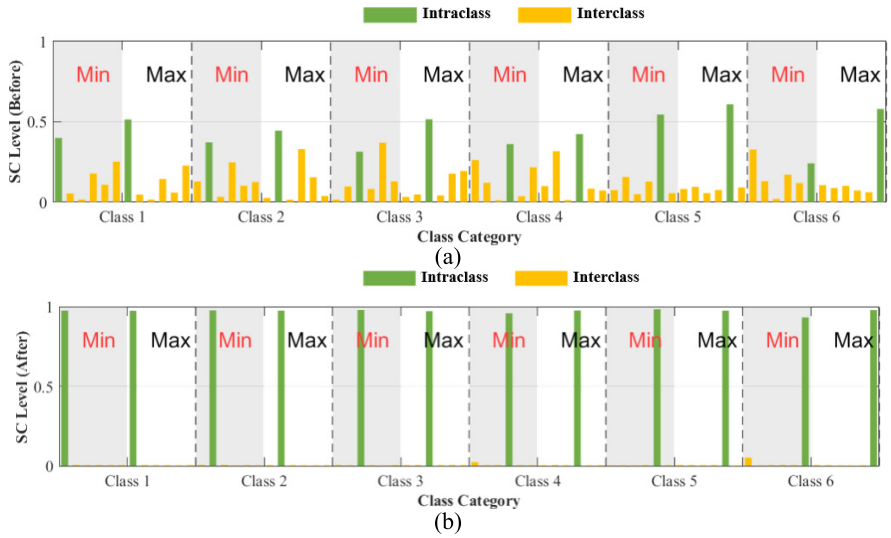


Fig. 3. Changes description of SC levels before and after optimization. For each class, take two samples in which the maximum and minimum belong as an example. There are six categories in this example. (a) Before. (b) After.

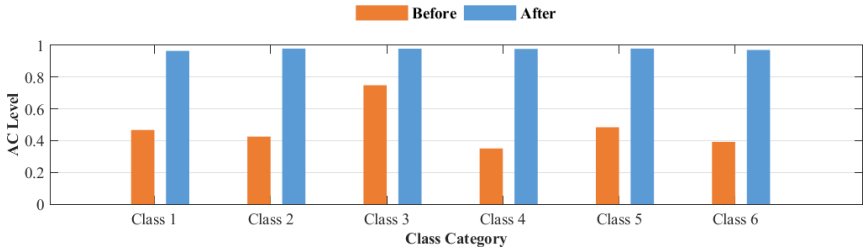


Fig. 4. Corresponding changes of AC levels of each class in Fig. 3.

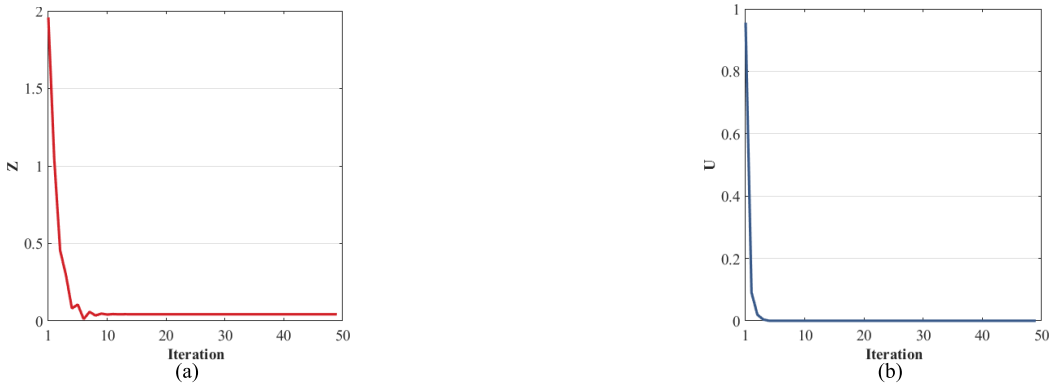


Fig. 5. Convergence of function (13) and (20). (a) Value of Z. (b) Value of U.

The coefficient is updated by

$$A_{ij}^{t+1} = A_{ij}^{t} - s^{t}\frac{\partial F1}{\partial A_{ij}} \tag{17}$$

where s^t is the step size and A_{ij}^t and A_{ij}^{t+1} are the current and new values, respectively. Equation (18) is considered as the objective function to decide whether to accept the updated value or not and adjust the step size

$$Z = \left|\left|A_{ij}^{t+1} - \tilde{A}_{ij}\right|\right|_2^2. \tag{18}$$

A_{ij}^{t+1} will be accepted if it results in lowering the function value. The current step s^t will be updated according to

$$s^{t+1} = \begin{cases} 2s^t, & Z < \eta \\ s^t, & Z = \eta \\ 0.5s^t, & Z > \eta \end{cases} \tag{19}$$

where η is the threshold. Fig. 5(a) shows the process of convergence.

It can be seen from Figs. 3 and 4 that the discrimination has been substantially improved. Before optimization, SC levels of all classes are around 0.5, while they can be effectively raised

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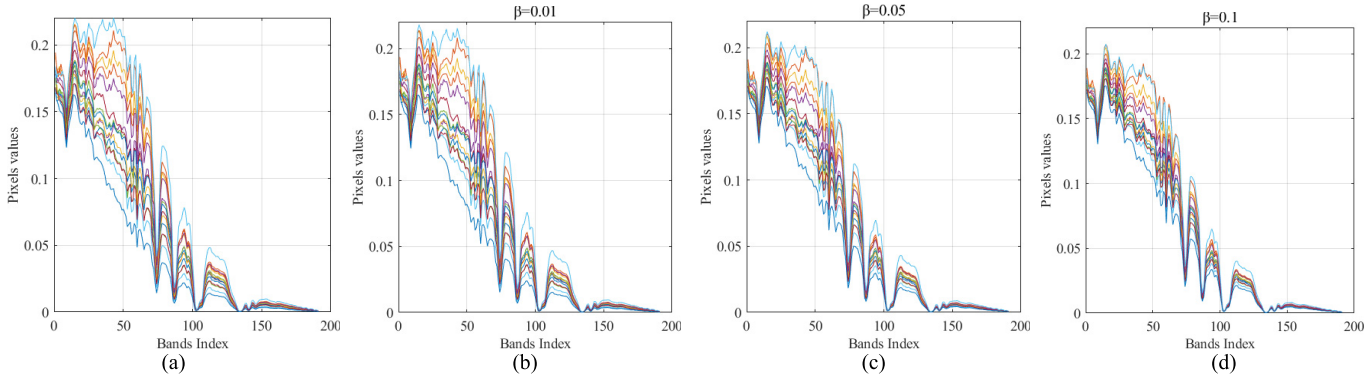


Fig. 6. Distribution of spectral bands with different balance parameters β. It can be seen that the factor can adjust the distance between samples, which represents the similarity and difference while leaving the characteristics in sensitive bands unaffected. (a) Original spectral band information. (b) Updated spectral bands information ($\beta = 0.01$). (c) Updated spectral bands information ($\beta = 0.05$). (d) Updated spectral bands information ($\beta = 0.1$).

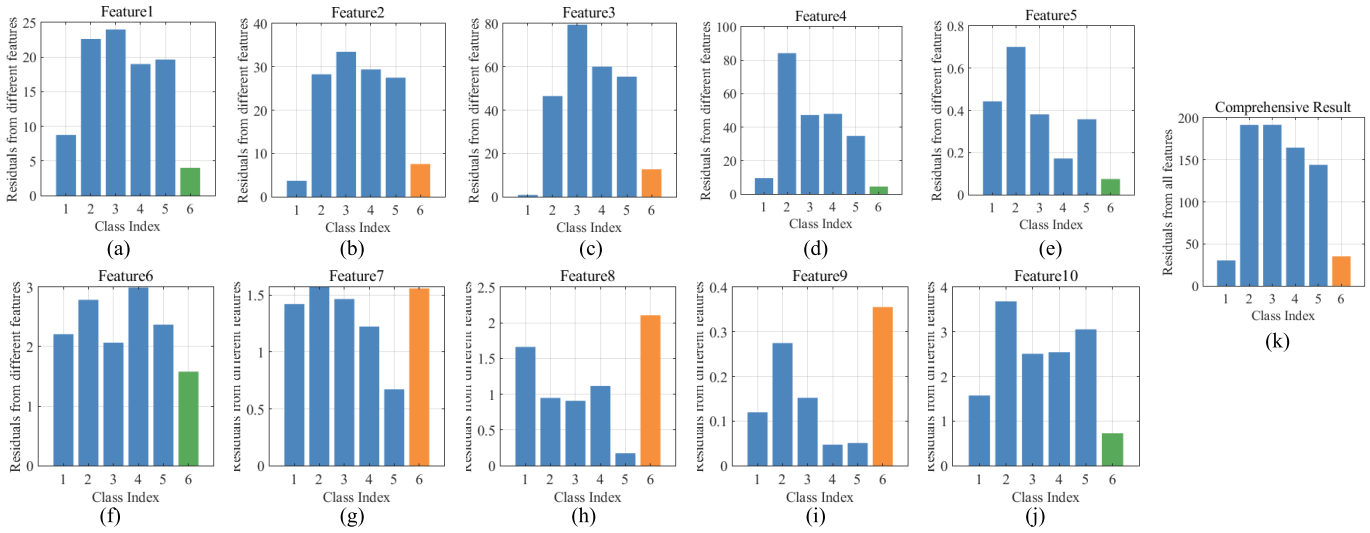


Fig. 7. Reconstruction errors from different features. The larger residual values can reach nearly hundreds and the small one is only a few tenths. Their huge difference in the order of magnitude of residuals can lead to misclassifying. (a), (d), (e), (f), and (j) Right. (b), (c), (g), (h), and (i) Wrong. (k) Wrong result [add up different errors according to (10)].

above 0.9 or even close to 1 after an optimizing process. It also applies to the improvement of AC levels, as shown in Fig. 4. Nevertheless, the reconstruction power should not be ignored. Thus, a balance program is implemented again. The objective function can be written as

$$F2 = \arg\min_{D_l} ||D_l A_l - X_l||_2^2 + \lambda ||A_l||_2^2 + \beta \omega_l ||D_l - \bar{D}_c||_2^2 \quad (l = 1, 2, \ldots, Nc) \tag{20}$$

where $X = D$ is the items to be updated, A is the updated coefficient matrix according to the above optimization stage, D is the original dictionary, $\bar{D}_c$ represents the mean of similar samples from the cth class, ω is used to measure the different importance of similar samples, and β is to balance the difference and similarity between them. Fig. 6 shows the distribution of spectrum under different balance parameters. It can control the balance effectively. On the other hand, the diversity of the samples has been increased due to the consideration of the similarity and difference between samples.

In the dictionary balance stage, the objective function can have a closed solution, which can be calculated by

$$D_l^* = \left(X_l A_l^T + \beta \bar{D}_l\right)\left(A_l A_l^T + \beta \omega_l I\right)^{-1} \tag{21}$$

where D_l^* is the updated item. As for ω, it is obvious that the relationships between similar samples are uncertain. Thus, the entropy is introduced to determine weights by

$$\min_{\omega_l} \sum_{l=1}^{Nc} \beta \omega_l ||D_l - \bar{D}_c||_2^2 + \sigma \omega_l \ln \omega_l \tag{22}$$

where σ is the Lagrange multiplier. Then, the weights can be calculated by

$$\omega_l = \exp\left(-1 - \sigma ||D_l - \bar{D}_c||_2^2\right). \tag{23}$$

The convergence condition is

$$U = \left|\left|\omega^{t+1} - \omega^t\right|\right|_2^2 / ||\omega^t||_2^2 \tag{24}$$

where t is the number of iterations. Fig. 5(b) shows the convergence value.

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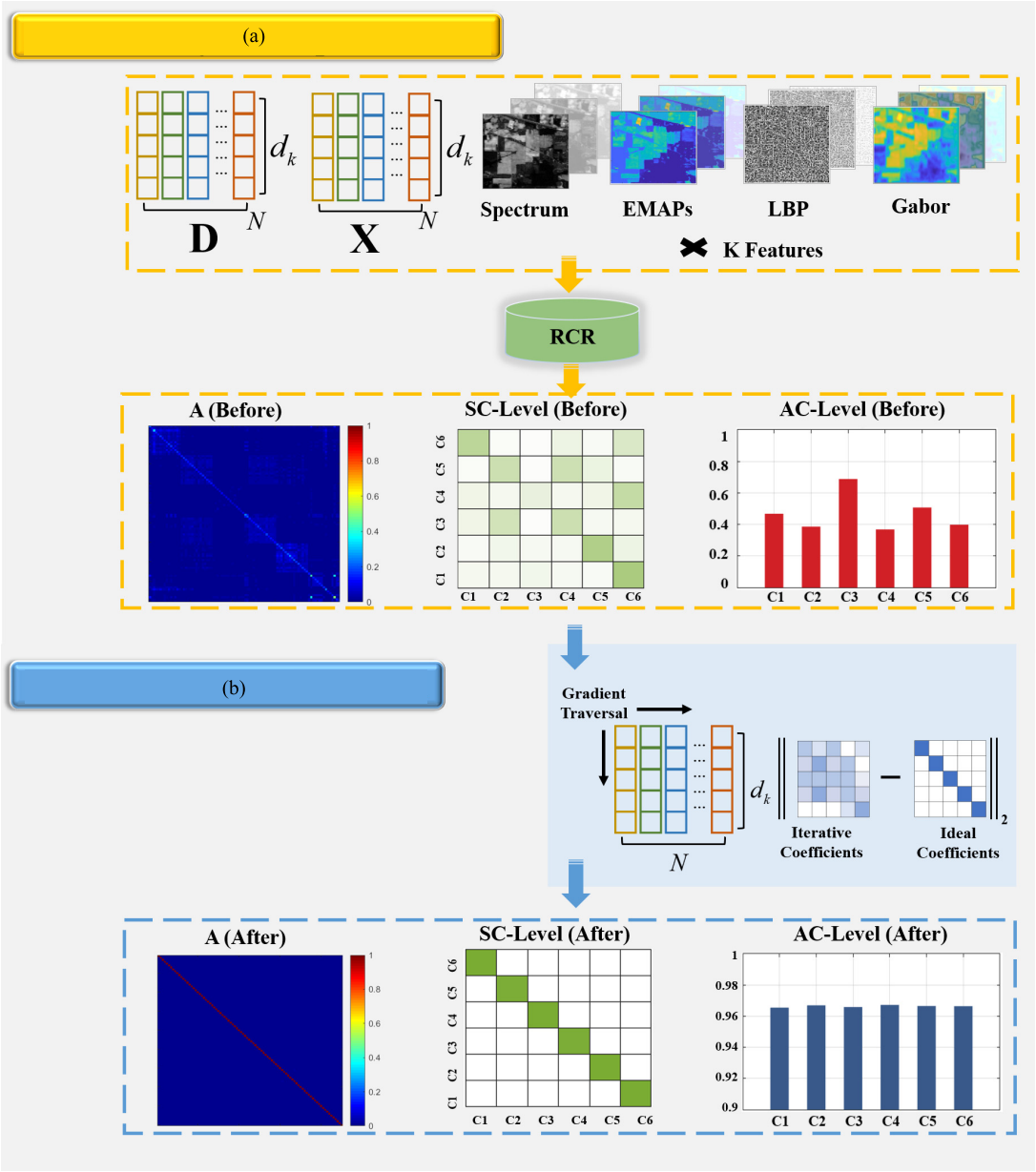


Fig. 8. Flow of self-representation and coefficients optimization stages. (a) Dictionary self-representation. (b) coefficients optimization.

After that stage of optimization and balance, an updated dictionary, which considers discrimination and sharedness simultaneously, is formed. Then, the learned dictionary can be input into RCR for classification.

Besides, the magnitude of residuals calculated by RCR is in huge different orders. The larger residual values can even reach nearly hundreds and the small one is only a few tenths, as shown in Fig. 7. Apparently, the large ones can influence the classification accuracy since the errors applied in label determination are summarized. To reduce the abovementioned effect, a new classification approach is designed without calculating the additive reconstruction errors.

In the label determination stage, the two designed indicators SC and AC are employed. It is supposed to be noticed that all the dictionary items will be applied to calculate a new contribution error. It can be described as

$$e_c^k = \frac{\left(P_c^k\right)^T \alpha^k}{Q_c^k} \tag{25}$$

$$E_c = \prod_{k=1}^{K} e_c^k \mathrm{var}\left(e_c^k\right) \tag{26}$$

$$\mathrm{class}(y) = \max(E_c) \tag{27}$$

where α^k is the coding coefficient obtained by (1). In this circumstance, testing samples are put into the RCR model to collect the corresponding representation coefficients α^k. The variance of various errors $\mathrm{var}(e_c^k)$ is the constraint term describing the dispersion of contribution levels from various features to weaken the influence of feature differences.

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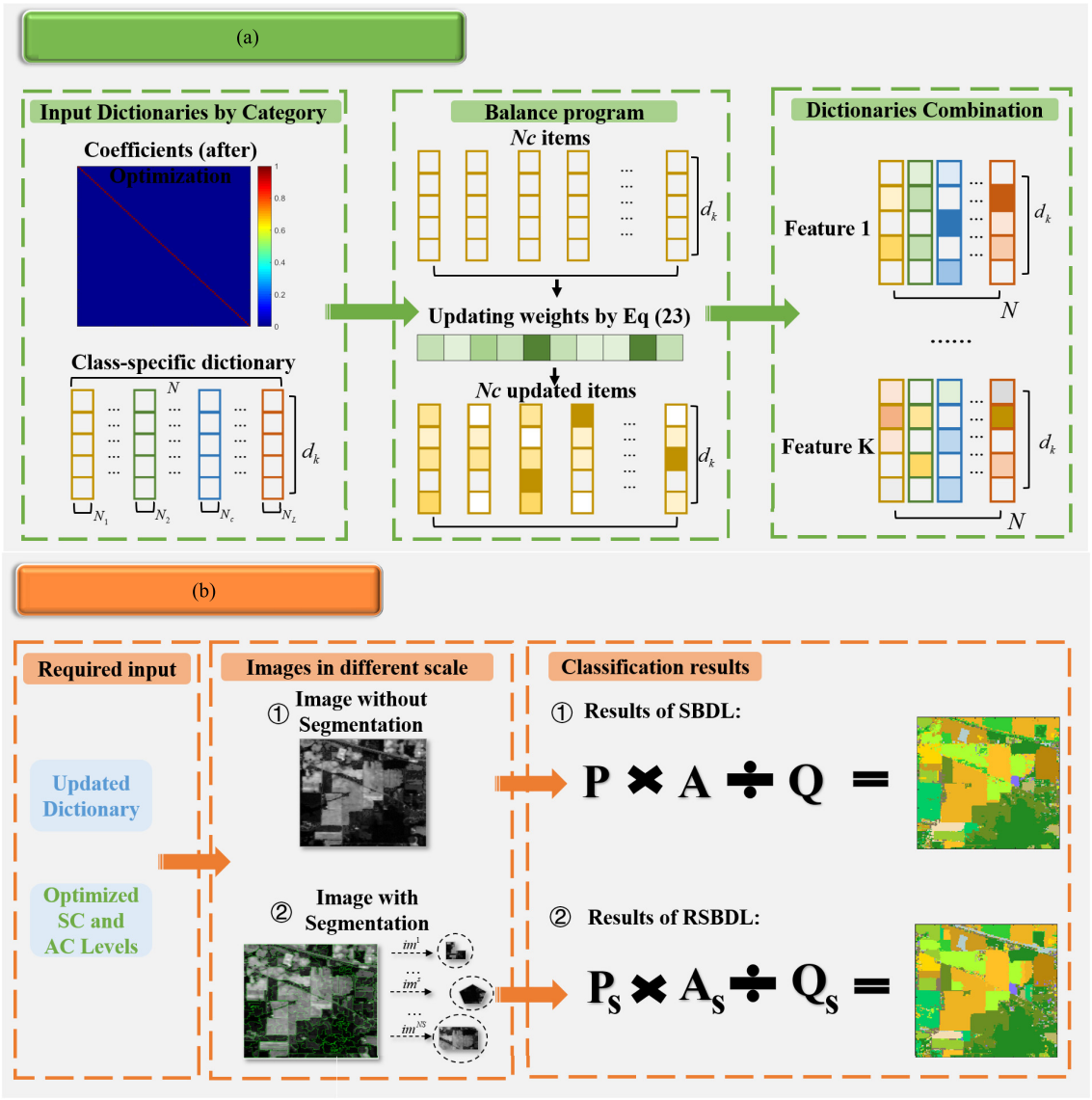


Fig. 9. Flow of self-balancing and labels determination stages. (a) Self-balancing dictionary. (b) Label determination.

The detailed introduction is given in Algorithm 1. The flow is shown in Figs. 8 and 9.

B. RSBDL_RCR

The proposed SBDL_RCR learns a balanced dictionary and shows its flexibility in the classification stage for applying all samples with SC and AC levels. However, its ability of discrimination improvement can be degraded with the complexity of image increasing. To respond to the problem, a region-based version is proposed as a solution. The hyperspectral image can be segmented first to reduce the number of classes within a subimage, and then, the proposed SBDL_RCR can work effectively on each segment.

In this version, image segmentation plays three significant roles in a classification accuracy improvement.

First, it can strengthen the capability of discrimination improvement introduced above. As shown in Fig. 10, it can be found that the SC can be improved from 0.1660 to 0.9695, 0.1549 to 0.9469, and 0.1406 to 0.8980 at least for three data which have six, nine, and 16 categories, respectively. It is obvious that with a number of classes increasing, the improved SC decreases. For example, for the most complex one among three data, the maximum of improved SC can just reach 0.9317, while the other two are 0.9709 and 0.9691. Through segmentation, the improved SC can reach 0.9991 in this circumstance. There are three datasets in different complexity being applied for comparison. Also, it can be seen that using image segmentation can effectively improve the discrimination further, especially for more complex data.

Second, it can reduce the computation cost. Fig. 11 compares the time of classifying the whole image, and it can be seen that although the time will be prolonged with the number of segments increasing, it is largely shorter than classification without segmentation, especially for bigger data. Indian Pines, Purdue Campus, and Yancheng Wetland (Dafeng Natural Reserve) data have 21 025, 193 024, and 623 152 pixels, respectively.

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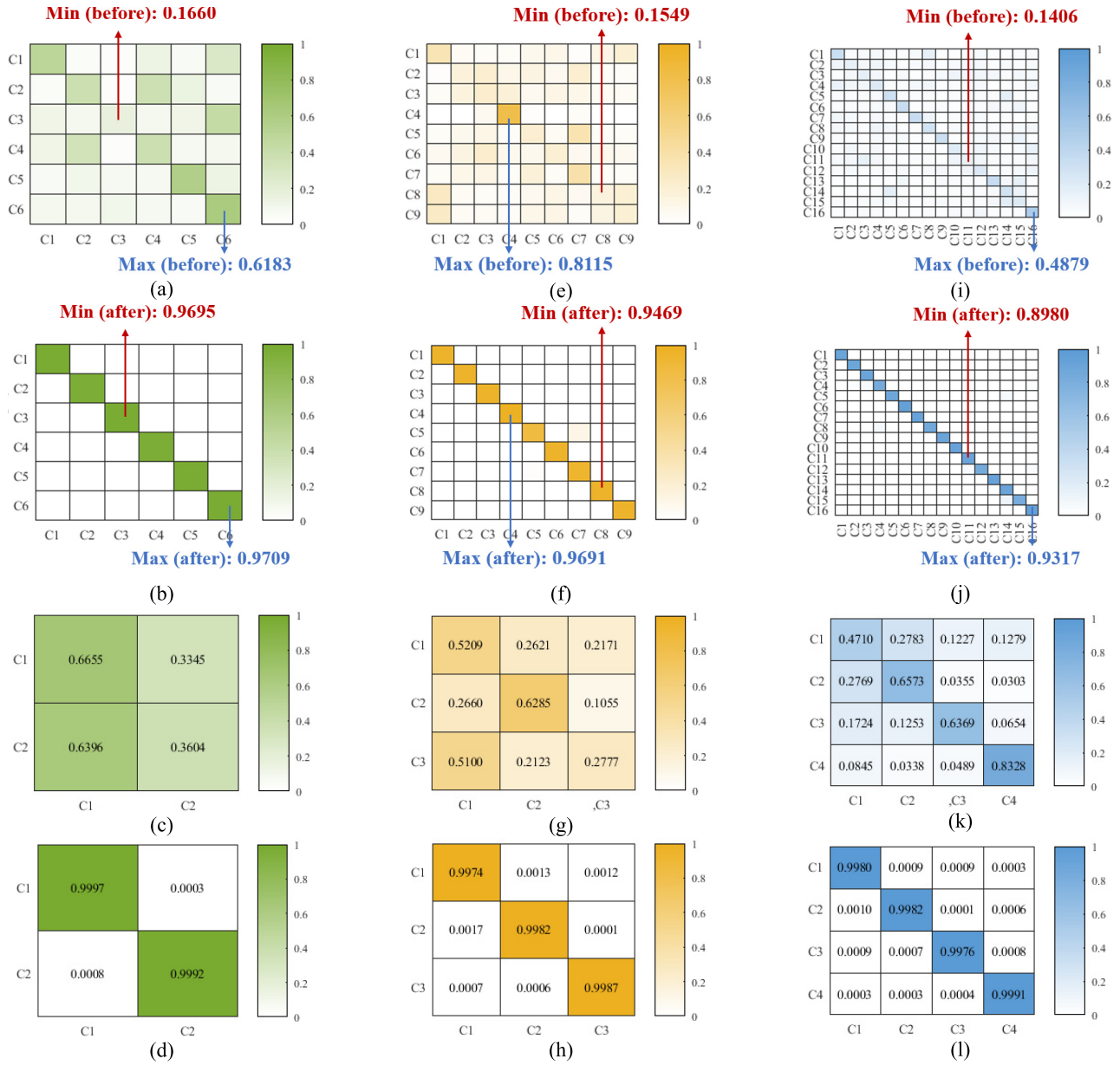


Fig. 10. Comparison of SC level in different complexities. (a)–(d), (e)–(h), and (i)–(l) Results of SC changes after segmentation, and the three data include six, nine, and 16 categories, respectively. Due to segmentation, the categories in some subimages can be 2, 3, and 4. It is obvious that segmentation can help to improve the discriminability further.

Third, it can effectively improve the classification performance. Fig. 12 shows the overall accuracy (OA) of three datasets to introduce the relationship between the number of training samples per class and subimages. It can be seen that with the number of subimages increasing, the accuracy increases in different degrees. The increasement is larger for more complex data (e.g., Indian Pines dataset has 16 classes).

The detailed procedures of RSBDL_RCR are summarized in Algorithm 2.

IV. Experiments

A. Dataset

1) Purdue Campus (PD): The third experimental data are collected by the HyMap system over the target area. An image with about 3.5 m/pixel is regarded as our experimental scene to be classified, which has 377 × 512 pixels with 126 bands. Although there are a few types of classes, the number of all samples is large, which causes a longer time for classification

TABLE I
Six Ground-Truth Classes of the HYMAP Dataset

No	Class	Samples
C1	Road	1287
C2	Grass	1114
C3	Shadow	219
C4	Soil	379
C5	Tree	1351
C6	Roof	1285
Total		5635

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TABLE II
NINE GROUND-TRUTH CLASSES OF THE GF-5 DATASET

No	Class	Samples
C1	Sea	1020
C2	Mud flat	528
C3	Road	150
C4	Building	89
C5	Farmland	918
C6	River	276
C7	Tree	724
C8	Acquaculture	746
C9	Lake	79
Total		4530

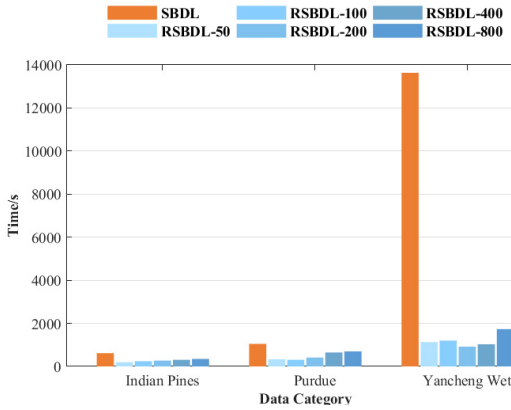


Fig. 11. Time spent over three data with and without using segmentation (nBs = [50 100 200 400 800]).

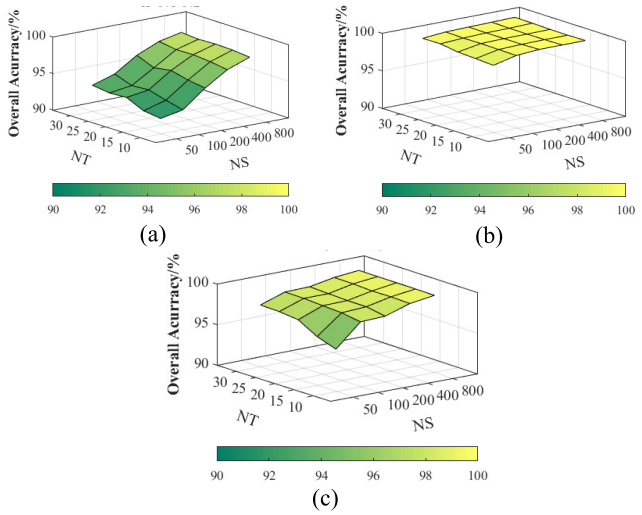


Fig. 12. Relationship between the number of subimages, training samples, and OA. (NT: number of training samples per class and NS: number of subimages. (a) IP, (b) PD, and (c) YC).

compared with the other two datasets. Table I gives the information of labeled samples.

2) Dafeng Natural Reserve (Yancheng Wetland, YC): The second experimental data are acquired by GF-5 with 256 bands [65]. The experimental scene contains 986 × 632 pixels and nine different classes. As a typical coastal wetland, it is located in Yancheng, Jiangsu, China. The sea, farmland, and aquaculture occupy most of the area and distribute regularly. The detailed information of the labeled samples is shown in Table II.

TABLE III
SIXTEEN GROUND-TRUTH CLASSES OF THE AVIRIS DATASET

No	Class	Samples
C1	Alfalfa	46
C2	Corn-notill	1428
C3	Corn-mintill	830
C4	Corn	237
C5	Grass-pasture	483
C6	Grass-trees	730
C7	Grass-pasture-mowed	28
C8	Hay-windrowed	478
C9	Oats	20
C10	Soybean-notill	972
C11	Soybean-mintill	2455
C12	Soybean-clean	593
C13	Wheat	205
C14	Woods	1265
C15	Buildings-Grass-Trees-Drives	386
C16	Stone-Steel-Towers	93
Total		10249

TABLE IV
OPTIMAL PARAMETERS RESULTS

Method	Data	λ	τ	γ	β	σ
SBDL -RCR	PD	1.00E-04	1.00E-01	5.00E-01	5.00E-01	1.00E-01
	YC	1.00E-05	1.00E-02	1.00E-02	1.00E-02	1.00E-01
	IP	1.00E-05	1.00E-02	1.00E-02	5.00E-02	1.00E-01
RSBDL -RCR	PD	1.00E-01	1.00E-02	1.00E-01	1.00E-01	1.00E-01
	YC	1.00E-04	1.00E-01	1.00E-01	1.00E-01	1.00E-01
	IP	1.00E-04	1.00E-02	5.00E-02	5.00E-01	5.00E-02

3) Indian Pines (IP): The Indian Pines data were captured by AVIRIS sensor with the resolution of 20 m/pixel. The image contains 145 × 145 pixels over an agricultural area in NW Indiana, USA. The number of different surface materials is 16. After removing 20 disturbed by water absorption bands, the remaining 200 bands are the actual bands used in our experiment. Table III shows the information of labeled samples.

B. Classification

1) Experimental Setup: Since the proposed methods are mainly about an improvement of the representation learning method, some related state-of-the-art representation methods, such as CRC [22], TCRC [14], RCR [44], and MCRC_DL [42], are used for comparison. The classical machine learning method SVM is compared with. Since the

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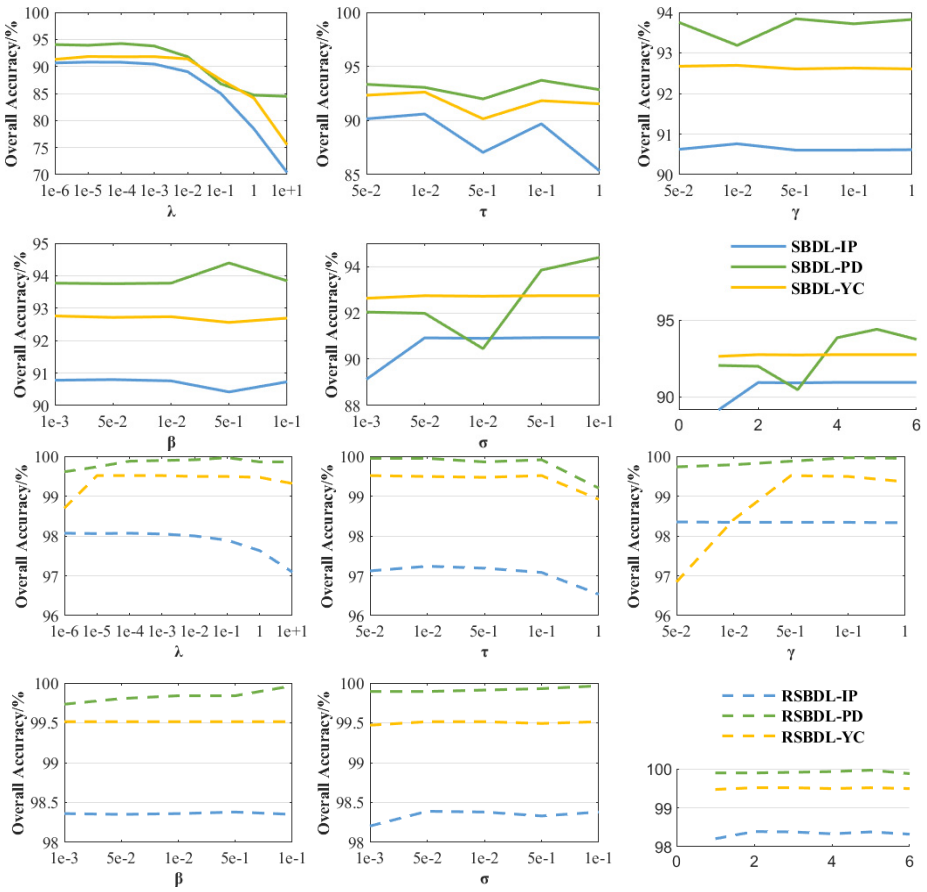


Fig. 13. Results of parameters analyzation.



Fig. 14. Results of OA, including the comparison of two label determination strategies. (a) Indian Pines (IP). (b) Purdue Campus (PD). (c) Dafeng Natural Reserve (YC).

spatial information is also considered, the SVM-CK that extracts spatial features is included. Besides, two methods, which are not in the family of representation methods, LCTL [64] and IFRF [15], are included to prove the effectiveness of the proposed methods. In the stage of data preparation, the multiple features, including EMAPs, LBP, and Gabor features, are extracted, subsequently selecting equal samples from each class to gather an initialized dictionary. In the stage of parameters analyzation, the five parameters λ, τ, γ, β, and σ are carefully analyzed to find the optimal group and observe the sensitivity. In the classification stage, the comparison between our methods and the others is completed. Three indicators, including OA, average accuracy (AA), and kappa coefficients, are employed to evaluate the effectiveness. Besides, the result obtained by the way of reconstruction errors is necessarily compared with the proposed one, which is calculated by the contribution levels of all training samples.

2) Parameters Analyzation: Fig. 13 shows the analysis results of five parameters. Also, the optima are listed in Table IV. The regularization parameter λ is sensitive to SBDL_RCR but less sensitive to RSBDL_RCR. It is because the segmentation reduces the impact of the parameters. In this circumstance, each subimage shares the same parameter value. The scale parameters τ and γ are used to adjust the similarity and difference between features according to (3). Since the features are used to describe the same samples simultaneously,

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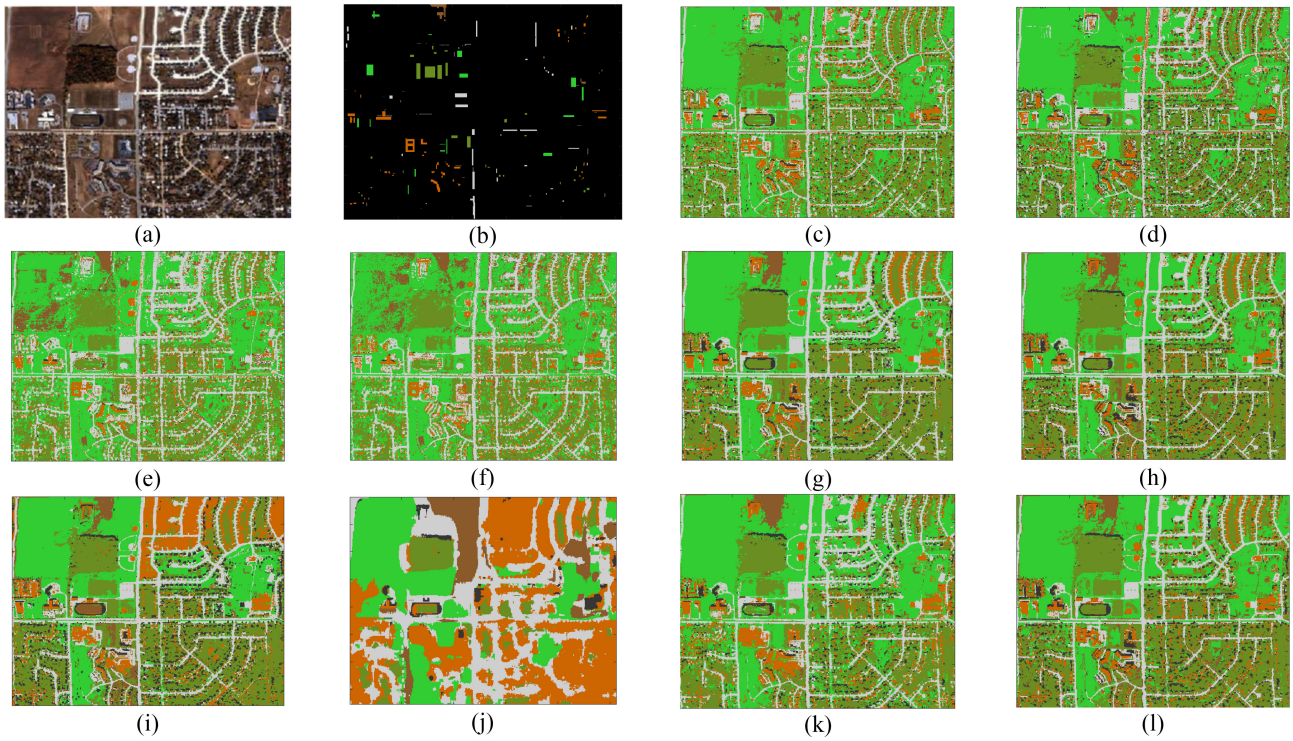


Fig. 15. Classification results of all samples from the HYMAP data. (a) False-color image. (b) Ground truth. (c) SVM. (d) SVM-CK. (e) CRC. (f) TCRC. (g) MCRC_DL. (h) RCR. (i) LCTL. (j) IFRF. (k) SBDL_RCR. (l) RSBDL_RCR.

TABLE V
CLASSIFICATION ACCURACY FOR THE PURDUE CAMPUS DATASET

Class	Train	SVM	SVM-CK	CRC	TCRC	MCRC-DL	RCR	LCTL	IFRF	SBDL-RCR	RSBDL-RCR
1	15	95.18	94.79	90.29	85.39	97.59	95.49	93.24	99.07	97.01	**100.00**
2	15	96.41	99.19	89.14	96.05	88.69	86.36	83.35	99.19	**99.27**	**100.00**
3	15	97.72	89.95	59.36	80.37	99.09	98.17	98.53	98.63	98.53	99.54
4	15	94.20	98.15	96.57	92.61	97.89	96.04	86.26	95.51	95.33	100.00
5	15	98.30	99.63	96.67	99.04	99.93	100.00	98.80	100.00	99.70	100.00
6	15	84.20	76.89	76.34	84.51	95.02	84.59	87.09	92.92	**90.55**	**99.38**
OA (%)		93.70	92.78	87.63	90.86	95.88	92.42	90.95	87.25	**97.66**	**99.84**
AA (%)		94.33	93.10	84.73	89.66	96.37	93.44	91.21	86.72	97.55	99.82
Kappa		0.9207	0.9092	0.8444	0.8856	0.9482	0.9050	0.8862	0.8392	0.9705	0.9980
Time (s)		5.48	38.24	207.82	145.24	3316.21	3376.78	8.43	12.92	1047.60	698.31

the weights will not be a huge difference, which causes the parameters to fluctuate in a small range. The balancing parameter β, which controls the contribution of discrimination and sharedness, changes not very obvious because of the high similarity between samples from the same class. Practically, it is regarded as a group with σ, which controls the weights between similar samples. It is noticed that either of the two indispensable parameters changes sensitively can cause an impact on the OA. It is intuitive to see that even a little change of σ can influence the classification performance since it adjusts the balance between very similar samples. For RSBDL_RCR, the accuracy is always in a high level with the changes of five parameters.

3) Classification Performance: It can be seen from Fig. 14 that the proposed methods obtain higher accuracy. Figs. 15–17 show the classification results of different methods.

For the first data, it can be seen from Table V that SBDL_RCR can improve the OA by 5.24% compared with RCR and RSBDL_RCR can further improve it by 2.18%. SBDL_RCR performs great in grass and roof, which can also be observed from Fig. 15. Furthermore, RSBDL_RCR can classify each class well.

For the second data, the two proposed methods obtain greater classification performance compared with other methods. However, RSBDL_RCR can improve limitedly over SBDL_RCR. It is because the known samples occupy a small part of the whole data. It can be seen that RSDBL_RCR shows that a great capability of identifying the ground materials for the AA is more than 99%.

For the third data, the OA obtained by SBDL_RCR can be improved approximately by 4% compared with RCR. Due to the numerous categories of these data, the proposed region-based version can further improve the

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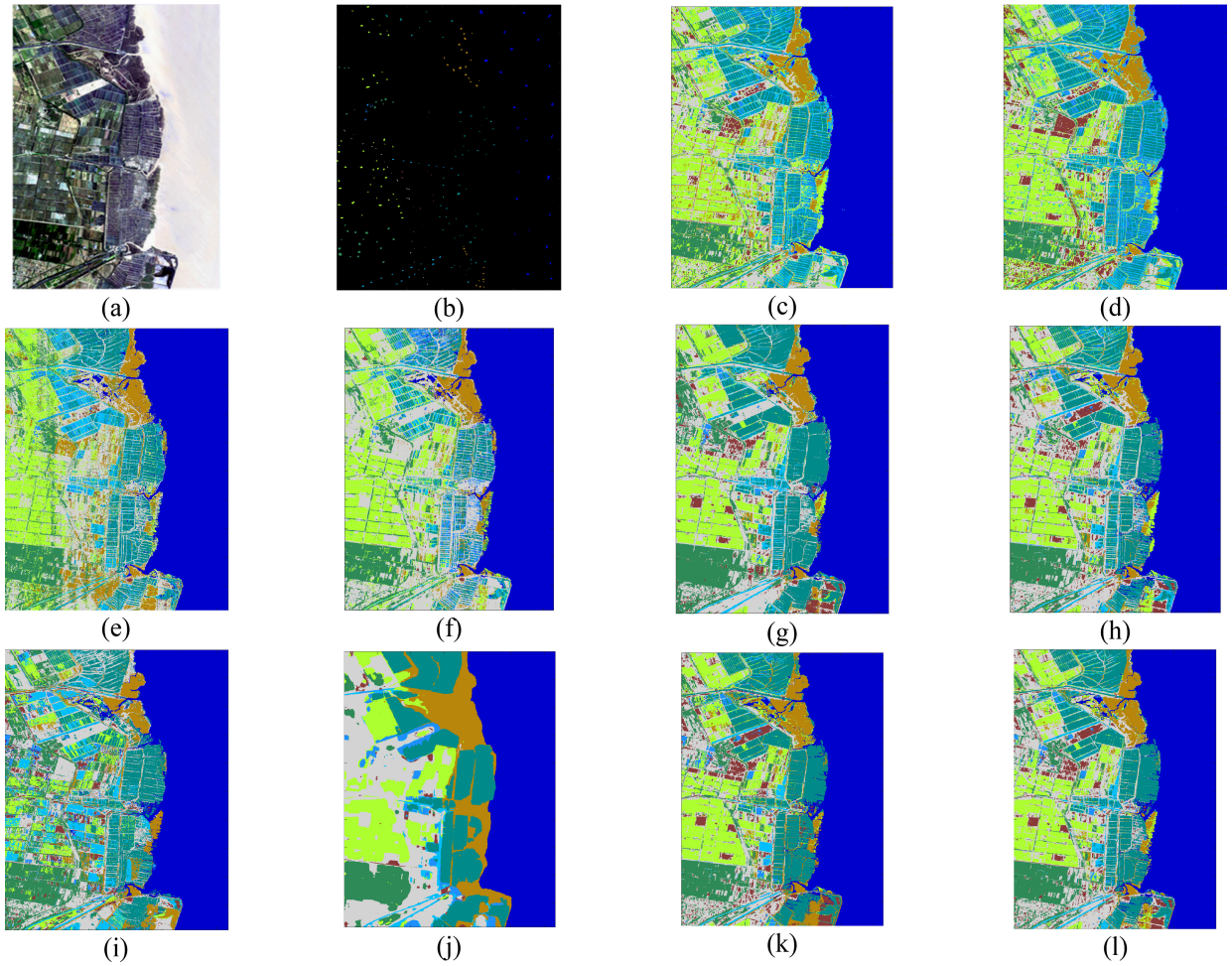


Fig. 16. Classification results of all samples from the GF-5 data. (a) False-color image. (b) Ground truth. (c) SVM. (d) SVM-CK. (e) CRC. (f) TCRC. (g) MCRC_DL. (h) RCR. (i) LCTL. (j) IFRF. (k) SBDL_RCR. (l) RSBDL_RCR.

TABLE VI
CLASSIFICATION ACCURACY FOR THE DAFENG NATURAL RESERVE DATASET

Class	Train	SVM	SVM-CK	CRC	TCRC	MCRC-DL	RCR	LCTL	IFRF	SBDL-RCR	RSBDL-RCR
1	15	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
2	15	79.73	80.87	94.32	81.06	84.99	92.44	86.55	99.81	**100.00**	**100.00**
3	15	92.00	97.33	86.00	82.67	96.30	32.54	65.19	26.20	97.33	99.33
4	15	100.00	100.00	93.26	98.88	98.65	88.10	86.49	100.00	100.00	100.00
5	15	79.19	92.48	78.87	89.65	86.82	87.27	42.08	98.07	95.32	98.26
6	15	68.84	84.06	54.35	51.09	98.85	95.74	84.29	88.53	92.03	**100.00**
7	15	87.57	91.02	92.13	89.64	98.03	98.23	96.19	95.08	**99.72**	**100.00**
8	15	85.66	81.10	78.28	67.69	98.63	97.75	88.51	100.00	100.00	99.73
9	15	86.08	93.67	86.08	92.41	100.00	94.12	100.00	78.05	100.00	100.00
OA (%)		86.67	90.53	86.60	85.01	94.78	90.01	81.77	89.51	**98.43**	**99.58**
AA (%)		86.56	91.17	84.81	83.68	95.81	87.35	83.26	87.30	98.27	99.70
Kappa		0.8419	0.8878	0.8409	0.8220	0.9380	0.8813	0.7850	0.8762	0.9813	0.9950
Time (s)		34.81	30.10	1824.31	2420.46	26575.53	13430.01	2.87	28.41	13632.82	1729.40

performance. Obviously, compared with SBDL_RCR, RSBDL_RCR can improve the OA by 5.84% again. It proves that the region-based version is effective for complex data. Nevertheless, the performance of some classes such as alfalfa and corn of RSBDL_RCR is not good as SBDL_RCR. It is because the subimages include part of the global information, which can be overcome by being segmented into an appropriate number of subimages.

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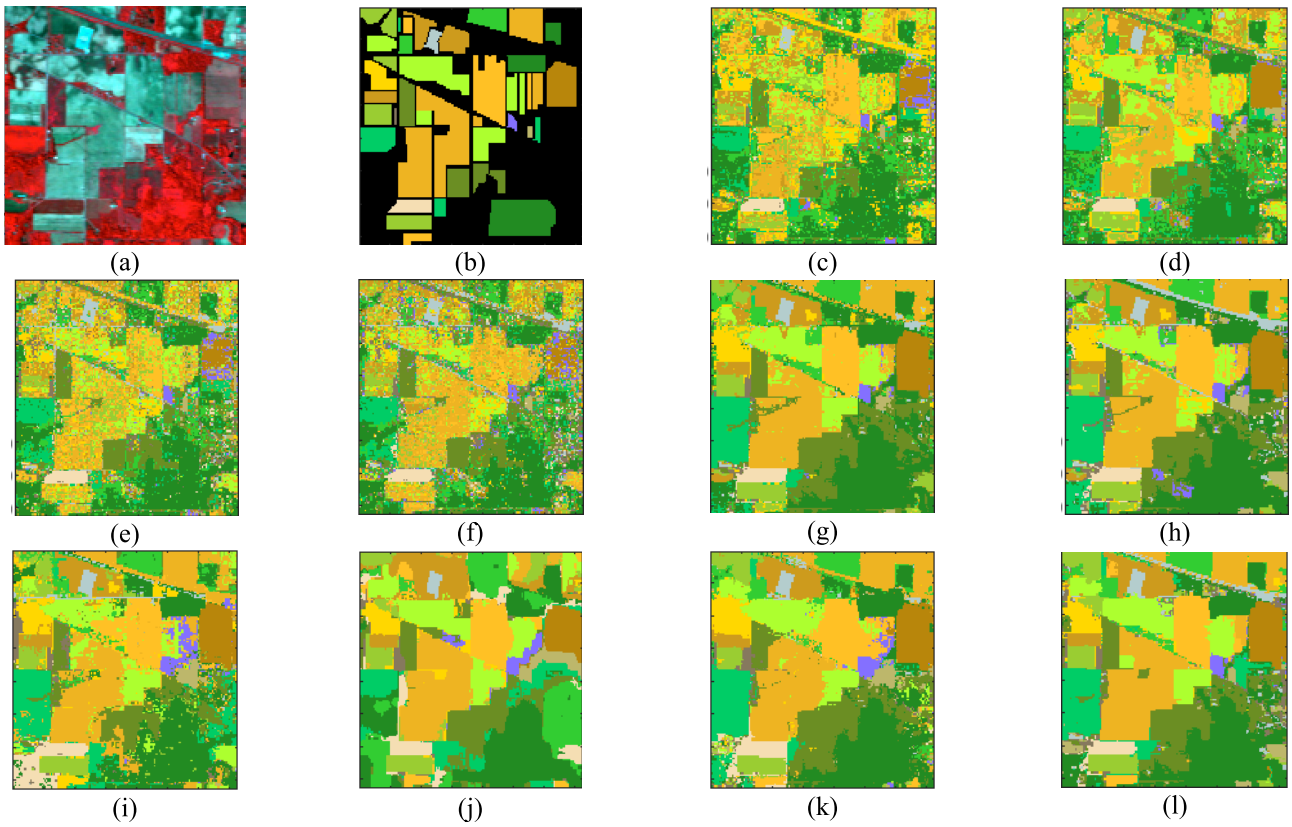


Fig. 17. Classification results of all samples from the AVIRIS data. (a) False-color image. (b) Ground truth. (c) SVM. (d) SVM-CK. (e) CRC. (f) TCRC. (g) MCRC_DL. (h) RCR. (i) LCTL. (j) IFRF. (k) SBDL_RCR. (l) RSBDL_RCR.*

TABLE VII
CLASSIFICATION ACCURACY FOR THE INDIAN PINES DATASET

Class	Train	SVM	SVM-CK	CRC	TCRC	MCRC-DL	RCR	LCTL	IFRF	SBDL-RCR	RSDBL-RCR
1	15	86.96	95.65	93.48	89.13	100.00	100.00	100.00	100.00	100.00	97.83
2	15	34.87	61.97	37.54	33.47	80.11	68.56	63.69	88.59	75.94	96.36
3	15	44.46	69.28	33.49	31.93	82.17	77.47	74.11	75.66	**86.01**	**98.80**
4	15	75.53	89.87	47.26	36.71	99.58	99.16	95.95	100.00	100.00	97.89
5	15	74.33	80.12	77.85	71.84	81.99	65.84	82.48	83.85	79.70	98.34
6	15	80.55	90.82	86.71	87.67	99.73	97.95	79.30	98.22	98.32	100.00
7	15	96.43	100.00	100.00	96.43	100.00	100.00	100.00	100.00	100.00	96.43
8	15	73.22	97.49	58.79	57.53	100.00	100.00	100.00	100.00	100.00	100.00
9	15	95.00	100.00	95.00	95.00	100.00	100.00	80.00	100.00	100.00	100.00
10	15	69.24	64.81	44.24	35.08	76.95	77.78	71.37	86.32	**88.51**	**89.30**
11	15	38.00	53.77	25.09	45.13	90.55	92.38	84.75	93.52	90.00	**98.53**
12	15	47.55	65.43	40.13	43.34	91.23	90.73	85.99	93.42	**95.67**	**99.49**
13	15	94.15	99.02	97.07	96.59	99.51	99.51	97.37	99.51	100.00	99.51
14	15	89.09	95.18	89.25	84.03	99.76	99.53	92.32	97.55	**98.80**	**100.00**
15	15	42.23	66.06	41.19	45.60	99.22	95.85	90.03	99.48	99.73	99.74
16	15	83.87	97.85	88.17	87.10	100.00	100.00	100.00	100.00	100.00	100.00
OA (%)		57.35	71.93	50.34	52.72	89.88	87.33	81.73	88.70	**91.99**	**97.83**
AA (%)		70.34	82.96	65.95	64.79	93.80	91.55	87.33	83.05	94.76	98.26
Kappa		0.5267	0.6854	0.4526	0.4720	0.8852	0.8564	0.7930	0.8715	0.9089	0.9753
Time (s)		13.46	7.87	42.56	90.73	1525.74	1226.28	1.92	6.07	617.24	346.81

According to Figs. 14–17, it can be seen that the accuracy decreases with the increasement of training samples when it reaches its maximum for Indian Pines data. The reason for this phenomenon is that the effectiveness of the proposed SBDL_RCR may be degraded when the data are in high complexity. Nevertheless, the region-based version can overcome this drawback to some extent. For the other two datasets, the accuracy will first increase to reach a high level and then

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maintain stably. It is noticed that the results obtained by the proposed contribution levels to classify perform better than the way of calculating the reconstruction errors.

In conclusion, SBDL_RCR and its region-based version perform well on three datasets. Compared with SBDL_RCR, the region-based version can further improve the accuracy of classification, especially for more complex data such as Indian Pines among all employed experimental datasets.

4) Complexity Analyzation: Tables V–VII list the cost of time from different methods. Compared with nonrepresentation learning methods such as SVM, LCTL, and IFRF, our methods have longer time since the compared one has higher efficiency with the help of C++ software and they employed spectral feature merely. As for other representation learning methods, CRC and TCRC have the less time consumption since they use only spectral information, while others use multiple features. The time complexity of the two methods is $O(n^3)$. It can be seen that our methods obtain higher accuracy but spend less time compared with MCRC_DL, which also joint dictionary learning and representation learning. In order to observe the computation cost, the space complexity is calculated either. For the objective function $F1$, the complexity is $O(d \times N \times N) + O(d \times N \times N) + O(d \times N \times N) = O(d \times N \times N)$, where d is the dimension. For gradient communication, the complexity is $O(d \times N) + O(d) + O(1) = O(d \times N)$. For the objective function $F2$, the complexity can be calculated as $O(d \times N) + O(N) + O(d \times N) = O(d \times N)$. Besides, RSBDL_RCR consumes less time than SBDL_RCR because the process of image segmentation reduces the complexity and speeds up the process. For RSBDL_RCR, the complexity can be reduced to $O = \max\{O(d \times Ns \times Ns)\}$, where Ns is the index of subimage. The results show that our proposed methods are competitive in dealing with images in different complexities compared with other classifiers.

V. Conclusion

In this article, a dictionary learning method is proposed to balance the power of discrimination and reconstruction between training samples. First, two evaluation indicators are proposed to measure the discrimination of self-representation training samples and are then optimized. Second, a balance program is imposed to adjust the contribution of similarity and difference to form a self-balancing dictionary. Third, a new framework is designed to determine the class labels without calculating the reconstruction errors. Finally, considering the decreased effectiveness when the complexity of data increases, a further version based on regions is proposed. The experimental results demonstrate that both of our methods can effectively improve the classification performance compared with other classifiers. There are still some limitations.

The process of optimizing the coefficients in the dictionary learning stage by using gradient descent is time consuming. It should be developed and replaced with some advanced methods such as AdaGrad and its variants.

The strategy designed for considering the difference between subdictionaries is not comprehensive enough. In the future, it may need a discriminative term to be added to the objective function to refine the approach.

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